

Mid-Semester Exam: Discrete Mathematics (CSE121)

Date: 27-09-2025

Maximum marks: 60

Instructions. Please read the following points carefully.

- Do not use your mobile phones, tablets, laptops, books, notebooks, or any other material during the exam.
- Write with either a black pen or a blue pen. Use of pencil is strictly prohibited.
- If you are found copying or talking to any other student, you will be disqualified from this exam, and the matter will be immediately reported to the Disciplinary Action Committee (DAC). If you have any questions, please ask the instructor or the invigilators.
- This exam is designed for 2 hours and 30 minutes.
- You can use the results given in the appendix without proving them. Apart from this, you can not assume any claim or result proven in the class/tutorial/quiz/homework. In such a case, you will have to prove things from scratch.

1 Problem Set

Partial marks for incomplete answers and missing steps depends on the solution. You are not allowed to bargain marks for incorrect and complete solution. And, it shall be considered as frivolous request, penalty will be imposed accordingly.

If your answers are not written in the form of given rubric then your answer is checked accordingly. It's not possible to make a rubric for all the solution.

Question 1.1. 1. (3 marks). Let p, q, r be propositional variables. Write the truth tables of the following propositions:

$$(a) p \implies (\neg q \oplus r) \iff (\neg p \vee q).$$

Rubric:

- If all the rows are correct 1.5 marks
- If any row is incorrect -0.25 marks

$$(b) p \implies \neg q \vee r \wedge \neg p \iff \neg r.$$

Rubric:

- i. If all the rows are correct 1.5 marks
 - ii. If any row is incorrect -0.25 marks
2. (3 marks). Convert the following English statements first to logical expressions, and then write their negations first in logic and then in English. You should mention the rule of first-order logic you are using at every step. The algebra of propositional logic is given in the appendix.
- (a) There exists a student of IIIT Delhi who knows singing or dancing.
 - (b) Every Indian citizen enjoys fundamental rights and bears fundamental duties.

Rubric:

- (a) Logical expression correctly formed from English 1 mark
- (b) Negation written correctly in logic with rule mentioned (De Morgan, quantifier negation, etc.) 1 mark
 Partial credit if rule not named but applied correctly: -0.25
- (c) Negation translated back to clear English 0.5 mark
- (d) Mention of rules of logic (quantifier negation, De Morgan's law, implication rule, etc.) 0.5 mark

Common Mistake deduction:

- Negation only in words, not logic: -0.5
- Rule missing but logic correct: -0.25
- Small English phrasing issues: no marks cut if meaning is correct

3. (2 marks). Let x, y be variables. Are $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$ equivalent for every predicate $P(x, y)$ and for every domains of x and y ? Justify your answer.

Rubric:

- (a) 0.5 marks: Correctly states they are not equivalent in general.
 - (b) 0.5 marks: gives valid counterexample
 - (c) 0.5 marks: explains meaning of quantifiers clearly
 - (d) 0.5 marks: clear conclusion tied to reasoning
4. (2 marks). What do you understand by the inverse, converse, and contrapositive of an implication $p \implies q$? Which of these are equivalent to each other?

Rubric:

- (a) 0.5 marks: define inverse
- (b) 0.5 marks: define converse
- (c) 0.5 marks: define contrapositive
- (d) 0.5 marks: correct equivalence stated

Question 1.2. 1. (1.5 marks). Let $p_1, \dots, p_n$ be propositional variables for some non-zero natural number n . Prove or disprove the following: There exists a propositional formula u in $p_1, \dots, p_n$ such that u is a tautology. Justify your answer by using the algebra of propositional logic given in the appendix.

Rubric:

- (a) 0.5 marks: proving claim stating that it is true
 - (b) 1. marks: for correct justification and correct u formula with all propositional variable.
2. (2.5 marks). Let p, q, r be propositional variables. Convert $(p \oplus q) \iff \neg r$ to an equivalent conjunctive form (need not be a conjunctive normal form) using only the algebra of propositional logic given in the appendix. Mention the rule used at every step.

Rubric:

- (a) -0.5 marks: if the formula used in the solution is not present in the appendix and not proved.
 - (b) 0 marks: for any other method apart from the algebra of propositional logic
3. (6 marks). Let $p_1, \dots, p_n$ be propositional variables. Let u_1 and u_2 be the propositional formulas defined as follows

$$u_1 := (p_1 \vee \neg p_2) \wedge (p_2 \vee \neg p_3) \wedge \dots \wedge (p_i \vee \neg p_{i+1}) \wedge (p_{i+1} \vee \neg p_{i+2}) \wedge \dots \wedge (p_{n-1} \vee \neg p_n) \wedge (p_n \vee \neg p_1).$$

$$u_2 := ((p_1 \wedge p_2 \wedge \dots \wedge p_{n-1} \wedge p_n) \oplus (\neg p_1 \wedge \neg p_2 \wedge \dots \wedge \neg p_{n-1} \wedge \neg p_n)) \wedge (p_1 \vee p_2 \vee \neg(p_1 \wedge p_2)).$$

Are u_1 and u_2 equivalent? Prove or disprove your answer.

Rubric:

- (a) 2.5 marks: binary marking for correct analysis of u_1
- (b) 2.5 marks: binary marking for correct analysis of u_2
- (c) 1 mark: equivalence conclusion only if the above analysis is correct.

Question 1.3. 1. (5 marks). Let a, b, c, d be integers such that $a + b + c = d$. Prove that d is odd if and only if either exactly one of a, b, c is odd or each of a, b, c is odd.

Rubric:

- (a) 1 mark: Defines even/odd correctly and states domain of integer variable.
- (b) 1 mark: Represents a, b, c as $2k$ or $2k + 1$.
- (c) 1 mark: Simplifies $a + b + c$ correctly and identifies when it is odd.
- (d) 1 mark: Checks both possible cases clearly.
- (e) 1 mark: Writes final logical conclusion.

Missing integer domain or definitions: 0.5 each.

Using “odd + odd = even” without proof or example: 1.

Proves only one direction: max 3/5 marks.

2. (3 marks). State whether the following is true or false. Justify your answers.

- (a) Let x be a variable. Then, $x + 5 \geq 10$ is a proposition.
- (b) Let $P(x)$ be the predicate ' $x + 5 \geq 10$ ', where the domain of x consists of all odd integers. Then, the negation of $\exists x P(x)$ is the following: for every even integer x , $x + 5 < 10$.
- (c) For every prime number a, b satisfying $a < b$, there exists an odd natural number c such that $a < c < b$.

Rubric: 0.5 for correct answer stating. 0.5 for correct justification.

Question 1.4. Prove the following. Whenever you deal with integers, you are allowed to use only the properties of integers given in the appendix. You should mention which rule of integers you are using at every step in the proofs.

1. (3 marks). Let a be an integer. Prove that a^2 is odd if and only if $a + 3$ is even.

Rubric:

- (a) 1 mark: recognize that a^2 is odd iff a is odd.
- (b) 1 mark: forward proof
- (c) 1 mark: backward proof

2. (3 marks). Let a, b be arbitrary integers. If $a^2 + b^2 = 0$ then prove that $a = 0$ and $b = 0$.

Rubric:

- (a) 1 mark: for writing the property the square for any integer and sum of it are non-negative.
- (b) 1 mark: for proof
- (c) 1 mark: writing conclusion correctly

3. (4 marks). Let a, b, n be natural numbers. If $n = a \cdot b$ then prove that $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Rubric:

- (a) 4 marks: Fully correct, logically consistent, clear contraposition proof.
- (b) 3 marks: Minor algebraic or notation error but reasoning correct.
- (c) 2 marks: Partial understanding of contraposition, some correct steps but incomplete proof.
- (d) 1 marks: Mentions contraposition but no logical connection shown.
- (e) 0.5 marks: for Proof by Example.
- (f) 0 marks: Irrelevant or incorrect reasoning.

Question 1.5. 1. (4 marks). Suppose there are finitely many prime numbers $p_1, \dots, p_n$ for some non-zero natural number n . Prove or disprove the following statement: there exists a natural number b , where $b \neq 1$ and $p_1 \cdots p_n + b$ is a prime number.

Rubric:

- (a) 1 mark: for correctly identifying $p \implies q$ is false.
- (b) 1 mark: for each case. (zero, composite, prime)

2. (6 marks). Prove that every collection of six people contains either a club of three people or a group of three strangers.

The Definition of club and stranger is given in the appendix (see Definition 1).

Rubric:

- (a) 1 mark: Stating the assumption that person A_1 has met at least three other people.
 1 mark: Concluding that if any two of those three have met, it forms a club of three.
 1 mark: Concluding that if none of those three have met, they form a group of three strangers.
- (b) 1 mark: Stating the assumption that person A_1 has not met at least three other people.
 1 mark: Concluding that if all three of these "strangers" have met each other, they form a club of three.
 1 mark: Concluding that if any two of these "strangers" have not met, it forms a group of three strangers.

Question 1.6 (12 marks). For a natural number k , the notation $\sqrt[k]{2}$ means $2^{\frac{k}{2}}$.

Consider the collection of all numbers of the type $\sqrt[k]{2}$, where k is a non-zero natural number and k is a perfect square (Definition 2). Some elements of this collection are $\sqrt[1]{2}$, $\sqrt[4]{2}$, $\sqrt[16]{2}$, $\sqrt[36]{2}$, etc.

Are there finitely many irrational numbers in this collection? Prove or disprove your answer.

Solution:

Step 1. Understanding the Problem

We are given a collection:

$$A = \left\{ \sqrt[k]{2} : k \in \mathbb{N}, k \text{ is a perfect square} \right\},$$

where for any natural number k , the notation $\sqrt[k]{2}$ is defined as $2^{k/2}$. (Here, $\mathbb{N} = \{1, 2, 3, \dots\}$.)

Examples: - $\sqrt[1]{2} = 2^{1/2} = \sqrt{2}$ - $\sqrt[4]{2} = 2^{4/2} = 2^2 = 4$ - $\sqrt[16]{2} = 2^{16/2} = 2^8 = 256$

We wish to determine whether there are **finitely many irrational numbers** in A .

Step 2. Reformulating the Set A

Since k is a perfect square, we can write $k = n^2$ for some $n \in \mathbb{N}$. Then:

$$\sqrt[k]{2} = \sqrt[n^2]{2} = 2^{n^2/2}.$$

Thus, the set becomes:

$$A = \left\{ 2^{n^2/2} : n \in \mathbb{N} \right\}.$$

Define $a_n = 2^{n^2/2}$ for $n \in \mathbb{N}$. We will analyze the rationality of a_n .

Step 3. When is a_n Rational?

We examine the exponent $\frac{n^2}{2}$:

- If $\frac{n^2}{2} \in \mathbb{Z}$, then $a_n = 2^{\text{integer}} \in \mathbb{Z}$, hence rational. - If $\frac{n^2}{2} \notin \mathbb{Z}$, then $a_n = 2^{\text{non-integer}}$, and we will show that in this case a_n is irrational.

Now, $\frac{n^2}{2} \in \mathbb{Z}$ if and only if n^2 is even, i.e., n is even.

Let us prove two claims:

Claim 1: If n is even, then a_n is rational. (2 Marks)

Proof: Let $n = 2m$, where $m \in \mathbb{N}$. Then:

$$a_n = 2^{(2m)^2/2} = 2^{4m^2/2} = 2^{2m^2}.$$

Since $2m^2 \in \mathbb{Z}_{\geq 0}$, it follows that a_n is a positive integer, hence rational. *Claim 1 proved.*

Claim 2: If n is odd, then a_n is irrational. (2 Marks)

Proof: Let $n = 2m + 1$, where $m \in \mathbb{N}$. Then:

$$n^2 = (2m + 1)^2 = 4m^2 + 4m + 1,$$

$$\frac{n^2}{2} = 2m^2 + 2m + \frac{1}{2},$$

$$a_n = 2^{2m^2+2m+1/2} = 2^{2m^2+2m} \cdot 2^{1/2} = (2^{m^2+m})^2 \cdot \sqrt{2}.$$

Now, suppose for contradiction that a_n is rational. Since $(2^{m^2+m})^2$ is a positive integer (hence rational and nonzero), we have:

$$\sqrt{2} = \frac{a_n}{(2^{m^2+m})^2}.$$

The right-hand side is a quotient of two rational numbers (with nonzero denominator), so it is rational. Therefore, $\sqrt{2}$ is rational. **(2 Marks if proved $\sqrt{2}$ is irrational)**

However, it is a classical result that $\sqrt{2}$ is irrational. (A brief proof: if $\sqrt{2} = \frac{p}{q}$ in lowest terms, then $2q^2 = p^2$, so p^2 is even, hence p is even. Write $p = 2r$, then $2q^2 = 4r^2 \Rightarrow q^2 = 2r^2$, so q^2 is even, hence q is even, contradicting that $\frac{p}{q}$ is in lowest terms.)

This contradiction shows that our assumption that a_n is rational is false. Hence, a_n is irrational. *Claim 2 proved.*

Step 4. Infinitude of Irrational Numbers in A (6 Marks)

From Claim 2, for every **odd** $n \in \mathbb{N}$, the number a_n is irrational. The set of odd natural numbers is infinite (e.g., $n = 1, 3, 5, \dots$). Therefore, A contains **infinitely many irrational numbers**. **(Require Proof)**

Final Conclusion

There are **infinitely many irrational numbers** in the collection

$$A = \left\{ \sqrt[k]{2} : k \in \mathbb{N}, k \text{ is a perfect square} \right\}.$$

Hence, the answer to the question “Are there finitely many irrational numbers in this collection?” is **No**.

This completes the proof.

Another rubric:

1. 3 marks for proving there are infinitely many odd numbers.

2. 3 marks for proving there are infinitely many odd perfect square.
3. 3 marks for proving $2^{k^2/2}$ is irrational if k is odd.
4. 3 marks for proving collection has infinitely many irrational.

There are many proofs for a question. We have given a sample solution.

Appendix

A Definitions and results

Definition 1 (Club and strangers). *Let G be a group of people. If every pair of people in this group has met, then we call it a club. If every pair of people in G has not met, then we call it a group of strangers.*

Definition 2 (Perfect square). *An integer a is said to be perfect square if there exists an integer b such that $a = b^2$.*

Fact A.1. *Let a be a natural number which is not prime. Then, there exists a prime number b such that $b < a$ and $b \mid a$.*

B Important properties of integers

The following properties hold for every integers a, b, c .

1. **Closure under addition and multiplication:** $a + b, a \cdot b$ are integers.
2. **Associativity of addition:** $a + (b + c) = (a + b) + c$.
3. **Associativity of multiplication:** $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
4. **Commutativity of addition:** $a + b = b + a$.
5. **Commutativity of multiplication:** $a \cdot b = b \cdot a$.
6. **Distributive properties:** $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.
7. **Identities:** $0 \neq 1, a + 0 = a, a \cdot 0 = 0$, and $a \cdot 1 = a$.
8. **Additive inverses:** There exists a unique integer m such that $m := -1 \cdot a$ and $a + m = 0$. The inverse of a is denoted $-a$.
9. **Subtraction:** $a - b$ is defined as $a + (-b)$.
10. **Cancellation:** If $a \cdot b = a \cdot c$ and $a \neq 0$ then $b = c$.
11. **No zero divisors:** If $a \cdot b = 0$ then either $a = 0$ or $b = 0$.
12. **Signed product:** $-a \cdot b = (-a) \cdot b = a \cdot (-b)$.

C Rules of the algebra of propositional logic

Let p, q, r be propositional variables, and T, F denote true and false. The rules are stated below.

1. Rules for AND.

- (a) **Commutativity of AND:** $p \wedge q \equiv q \wedge p$.
- (b) **Associativity of AND:** $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$.
- (c) **Identity for AND:** $T \wedge p \equiv p$.
- (d) **Falsity for AND:** $F \wedge p \equiv p$.
- (e) **Idempotence for AND:** $p \wedge p \equiv p$.
- (f) **Contradiction for AND:** $p \wedge \neg p \equiv F$.

2. Rules for OR.

- (a) **Commutativity of OR:** $p \vee q \equiv q \vee p$.
- (b) **Associativity of OR:** $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
- (c) **Identity for OR:** $F \vee p \equiv p$.
- (d) **Truth for OR:** $T \vee p \equiv p$.
- (e) **Idempotence for OR:** $p \vee p \equiv p$.
- (f) **Validity for OR:** $p \vee \neg p \equiv T$.

3. Other rules.

- (a) **Double negation:** $\neg(\neg p) \equiv p$.
- (b) **Distributivity of AND over OR:** $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
- (c) **Distributivity of OR over AND:** $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- (d) **DeMorgan's laws:** $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$.